Cambridge (CIE) IGCSE Physics
Paper 6: Alternative to Practical (Set E)

Saturday 26 April 2025

Afternoon (Time: 1 hour 0 minutes)

Total marks

/ 40

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.

Scan here to mark your mock exam
or visit the mock exams landing page for this course

1 (a) A student is determining the acceleration of free fall g using a pendulum. Fig. 1.1 shows the pendulum. Fig. 1.2 shows one complete oscillation of the pendulum.

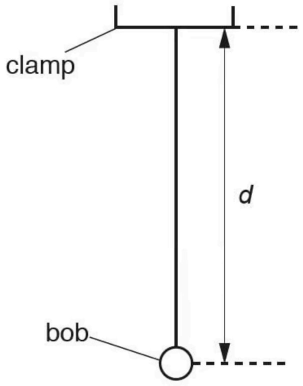

Fig. 1.1

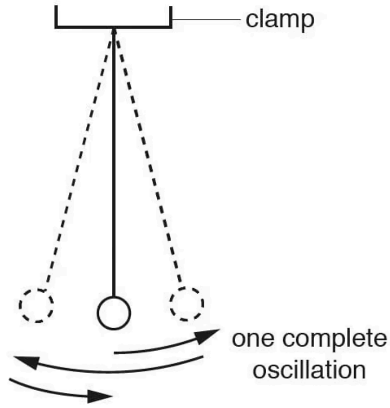

Fig. 1.2

On Fig. 1.1, measure the distance d.

d = ... cm

(1 mark)

(b) Fig. 1.1 is drawn 1/10th actual size. Assume that distance d from part (a) is 5.0 cm.

(i) Calculate the actual distance D from the bottom of the clamp to the centre of the bob.

D = ... cm [1]

The student displaces the bob slightly and releases it so that it swings. He measures the time t for 10 complete oscillations. The time t is shown on the stopwatch in Fig. 1.3.

Fig. 1.3

(ii) Write down the time t shown in Fig. 1.3.

t = ... [1]

(iii) Calculate the period T of the pendulum. The period is the time for one complete oscillation.

T = ... [1]

(iv) Calculate T^2.

T^2 = ... [1]

(v) Calculate the acceleration of free fall g using the equation $g = \dfrac{20}{T^2}$.

g = ... [1]

....

....

....

....

....

(5 marks)

(c) The student adjusts the pendulum until the distance D measured to the centre of the bob is 100.0cm.

He repeats the procedure and obtains another value of T^2.

T^2 = 3.92

(i) On the dotted line above, write the unit for T^2.

[1]

(ii) Calculate the acceleration of free fall g using the equation $g = \frac{40}{T^2}$ and the value of T^2 from (c). Give your answer to a suitable number of significant figures for this experiment.

g = ... [1]

....

....

(2 marks)

(d) Another student states that repeating the experiment improves the reliability of the value obtained for g.

Suggest **two** changes that you would make to improve the reliability. The stopwatch cannot be changed.

....

....

(2 marks)

(e) State **one** precaution that you would take in this experiment in order to obtain accurate readings.

....

(1 mark)

2 (a) A student is investigating a circuit containing different lamps. She is using the circuit shown in Fig. 3.1.

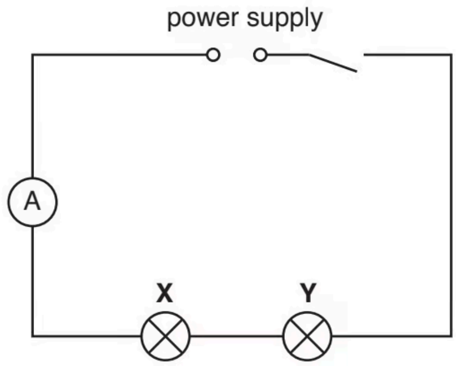

Fig. 3.1

On Fig. 3.1, draw a voltmeter connected so that it measures the potential difference (p.d.) across lamp **X**.

(1 mark)

(b) The student uses the ammeter to measure the current in the circuit.

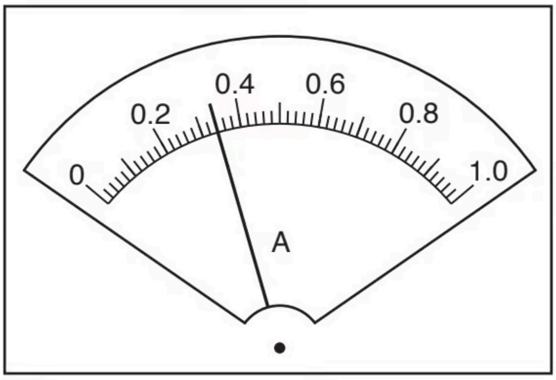

Fig. 3.2

Record the current I_S in amperes, A, in the circuit, as shown in Fig. 3.2.

I_S = ...

(1 mark)

(c) (i) The student uses the voltmeter to measure the p.d. V_X across lamp **X** and then reconnects the voltmeter to measure the p.d. V_Y across lamp **Y**.

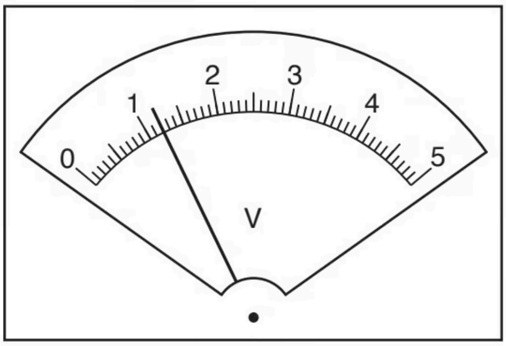

Fig. 3.3

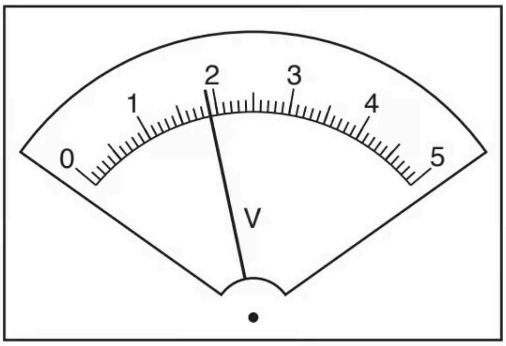

Fig. 3.4

Record the value of the p.d. V_X across lamp **X**, shown in Fig. 3.3.

V_X = ...

Record the value of the p.d. V_Y across lamp **Y**, shown in Fig. 3.4.

V_Y = ... [1]

(ii) She then measures the p.d. V_S across both lamps in series.

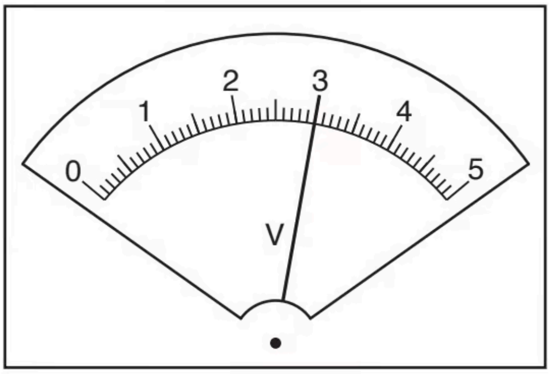

Fig. 3.5

Record the value of the p.d. V_S across both lamps in series, shown in Fig. 3.5.

V_S = ...[1]

(iii) A student suggests that V_S should be equal to (V_X + V_Y). State whether the readings support this suggestion. Justify your statement with reference to the results.

[2]

....

....

(4 marks)

(d) Calculate the resistance R_1 of lamp **X**. Use the readings from **(b)** and **(c)(i)** and the equation

$$R_1 = \frac{V_X}{I_S}.$$

R_1 = ... Ω .

(1 mark)

(e) (i) The circuit components are to be rearranged so that

- lamps **X** and **Y** are connected in parallel
- the ammeter measures the current in lamp **X** only
- the voltmeter measures the p.d. across the lamps.

Draw a circuit diagram of this arrangement.

[2]

(ii) The student sets up the circuit as described in **(e)(i)**.

She measures and records the current in lamp **X** and the p.d. across the lamps.

She then calculates a new resistance R_2 for lamp **X** in this parallel circuit.

$$R_2 = 8.3\ \Omega$$

The student notices that lamp **X** is very bright in this parallel circuit, but it was dim in the series circuit in **(a)**.

Suggest how temperature affects the resistance of a lamp. Justify your suggestion by reference to the value of R_1 from **(d)** and the value of R_2.

[2]

....

....

....

....

(4 marks)

3 (a) A student is investigating the reflection of light by a plane mirror.

Fig. 1.1 shows his ray-trace sheet at full size.

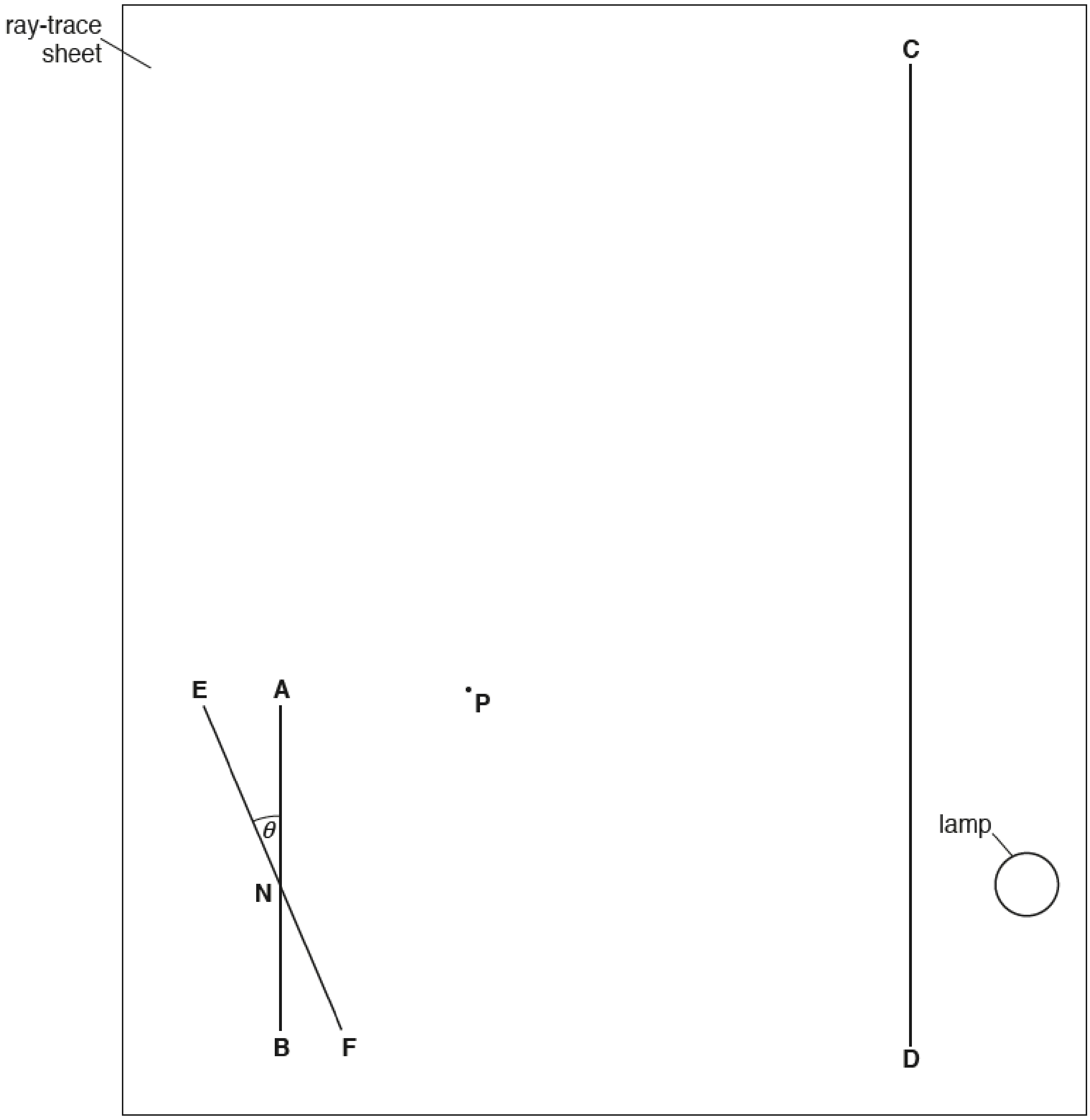

Fig. 1.1

The student carries out an initial experiment. He draws lines **AB** and **CD** as shown in Fig. 1.1. He then draws a line **EF** through a point **N** as shown in Fig. 1.1 and at an angle θ to line **AB**.

The angle θ is

measured to be 23° ±1°

Draw a normal to line **AB** at point **N** and extend the normal to line **CD**. Label the point at which the normal crosses line **CD** with the letter **L**.

....

(1 mark)

(b) The student places a plane mirror on line **EF** and a screen with a 2 mm slit on line **CD**. He arranges the screen so that a ray of light shines along line **LN**. The ray reflected from the mirror passes through point **P**.

State and explain whether point **P**, shown on Fig. 1.1, is at a suitable distance from point **N** for this investigation.

....

....

(2 marks)

(c) (i) Draw a line joining point **N** and point **P**. Extend this line until it meets line **CD**.

Label the point at which this line meets line **CD** with the letter **G**.

[1]

(ii) Measure the length *a* of line **LG**.

a = ... cm [1]

....

....

(2 marks)

(d) The student repeats the procedure for values of θ = 25°, 20°, 15°, 10° and 5°. His values for *a* are shown in Table 1.1.

Table 1.1

θ / °	a /cm
25	12.2
20	8.3
15	5.7
10	3.6
5	1.8

Use the values from Table 1.1 to plot a graph of a / cm (y-axis) against θ / ° (x-axis).

(4 marks)

(e) Suggest a possible source of inaccuracy in this experiment, even if it is carried out carefully.

(1 mark)

(f) A student wishes to check if his values for *a* are reliable.

Suggest how he could improve the experiment, using the same apparatus, to check the reliability of his results.

....

(1 mark)

4 A student investigates the effect of the colour of the surface of a metal container on the rate of loss of heat from the container. She knows that black surfaces are better radiators of thermal energy than white surfaces and wants to investigate the effect of other colours.

The following apparatus is available:

metal containers each with the outer surface painted a different colour

a thermometer

a stop-watch

a supply of hot water.

She can also use other apparatus and materials that are usually available in a school laboratory.

Plan an experiment to investigate the effect of the colour of the surface of a metal container on the rate of loss of heat from the container.

You should:

- draw a diagram of the apparatus used
- explain briefly how you would carry out the investigation
- state the key variables to be kept constant
- draw a table, or tables, with column headings, to show how you would display your readings (you are not required to enter any readings in the table)
- explain how you would use your readings to reach a conclusion.

(7 marks)